# DSCC/CSC/TCS 462 Final Project

Due: Tuesday, December 9, 2025 at 11:59 p.m. EST

## Introduction

This is the final group project for DSCC/CSC 462. You are expected to complete this project in groups of 3-4 people. Your group needs to submit a single file for all of you (i.e., you do not all upload the document; one group member does it, and that's sufficient). Only one person will upload it to their Blackboard at the assignment submission link, but please ensure that all group members' names are listed on the uploaded submission. All group members are expected to equally contribute to the final project, and you are responsible for knowing everything going on in the project (i.e., one person can't just be responsible for one question and not know what happened in the other questions; if I were to email you and ask you to explain to me what you group did for any questions, you should be very comfortable explaining the answer).

The project must be completed in RMarkdown. You will upload both your knitted PDF document and RMD file to Blackboard to be graded. The project writeup should be carefully completed: Please take the time to ensure your interpretations are thorough and thoughtful, and use the spellcheck feature to catch typos.

**Important**: Please label all graphs (title, axes, and legend, if applicable), and for any hypothesis tests, clearly state the hypotheses, test statistic, and p-value (or rejection region, if you choose to use it). You may use all `.test()` functions, i.e., no need to do any tests "by-hand." You must also use ggplot for graphics.

## Project Overview

The goal of this project is to **practice finding, cleaning, analyzing, and interpreting data**. There will be three main steps: (1) finding and cleaning data, and coming up with interesting questions to ask; (2) using tools from descriptive statistics to create visualizations and otherwise summarize the data; and (3) using tools from inferential statistics to run analyses and interpret results in the context of the questions posed in part (1). Guidance for all steps is presented below.

### Finding and Cleaning Data

The first step is finding data to analyze. Bear in mind that larger and richer datasets often lead to more interesting and variegated analysis but can also take more time to clean and visualize. As a general guideline, your data should consist of at least 10-15 different variables, each with at least 100-1000 observations. Please choose a dataset that interests your group; this will likely make the project much more enjoyable for all involved!

Here are some useful sources and guidelines for finding open-source data. Please feel free to reach out to the TAs or instructor if you have additional questions. Additionally, the librarians at UR are a wonderful resource; in particular, please contact Arjay Romanowski with any questions you have.

- The UR Library has a useful open dataset guide: (link).
- The US Federal Government releases data on data.gov: (link).
- Rochester has an open data portal: (link).
- New York City also has an open data portal: (link).
- US Census data is available: (link).
- ICPSR has a data repository: (link).

Keep in mind that once you have a dataset in mind, you will likely need to invest some effort in cleaning the data (i.e., making sure that it is properly formatted and without typos). For a (perhaps too-detailed) guide about how to clean data in R, check out the following resource: (link).

### Descriptive Statistics

Before making inferences, you must first summarize and visualize the data. You may approach this directive however you want, but please describe *why* you used the summaries and visualizations you did, as well as any *observations* you made based on the summaries or visualization. In addition, you *must* produce at least one form of output that fits each of the following formats, but it is more important to convey overall trends you observe in the data. Be conscious of your reader, and try to only include descriptions of the most important aspects of data (in particular, do not just plot everything you can, but rather make sure that your visualizations and descriptions help communicate useful information). Please note that this is a *minimum requirement*; often, you may find it useful to use more exploratory data visualizations to get a good sense of data.

1. Relative or absolute frequency table(s);
2. Relative or absolute frequency barplot(s);
3. Summaries of center (mean, median, mode, trimmed mean) and dispersion (IQR, variance, coefficient of variation, skewness);
4. Side-by-side boxplot(s);
5. Scatterplot(s), including correlation;
6. Histogram(s) with the appropriate number of bins and vertical lines corresponding to the mean and median;
7. Quantile plot(s);
8. Box-Cox transformation(s) (only if certain data looks non-normal).

### Inferential Statistics

Once you have visualized the data, it is time to start analyzing it. You must run at least one analysis of each of the following categories. For each analysis, pose it as a question to be answered at a specific significance level, and interpret the results accordingly. You may use hypothesis tests *or* confidence intervals, but you must use at least three hypothesis tests and three confidence intervals in your analysis overall. Again, please note that the following is a *minimum requirement* and that often you can answer more interesting questions by running more intricate tests. Lastly, please *do not feel pressure to find things statistically significant*! This is how bad science is done. Just ask questions about which you are genuinely interested in finding the answers, and interpret the data accordingly.

1. Inference about mean(s);
2. Inference about variance(s);
3. Inference about proportion(s);
4. Inference about two proportions;
5. $\chi^2$ inference (goodness-of-fit or test of independence);
6. ANOVA;
7. Inference about correlation;
8. Regression.

## Notes and Logistics

This is a chance to show off what you've learned over the course of the semester, so please be creative! Going above and beyond on the project is appreciated and may be rewarded with extra credit at the instructor's discretion. If you have any questions at any point, feel free to come to the instructor or the TAs, and we'll be happy to help.

There will be weekly project check-ins to make sure that you are actively thinking about this project. These

are low-stakes and will involve a short written submission, and you are encouraged to reach out to the TAs or instructor with any questions about the project. Additionally, there will be an opportunity to receive some extra credit by presenting your (perhaps not-yet-complete, but hopefully fairly close-to-done) project during class on December 4. Only exemplary projects will be chosen the previous week; more details will follow.